SIMATS ENGINEERING

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – CODING CHALLENGE QUESTIONS

Course Code:	CSA12	**Course Title:**	Computer Architecture
CO Assessed:	CO1 – Precision (BL4)	**Assessment:**	Coding challenge questions
Weightage:	40%	**Total Marks:**	100
CO1 Statement:	Analyse the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking		
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Section / Batch:	2023 batch	**Date:**	28/07/2026

Instructions to Students

- Analyze the given scenario and identify the problem requirements before developing the solution.
- Design the algorithm or flowchart and select the appropriate 8085 instructions, registers, and addressing modes required for implementation.
- Develop and execute the 8085 Assembly Language Program (ALP) using a simulator or trainer kit to solve the given problem.
- Verify and validate the output by testing the program with the given input data and ensuring the expected results are obtained.
- Interpret the program execution by explaining the logic used, the role of registers, memory locations, flags, and the final output generated.

QUESTIONS

1. A smart factory stores the hourly production count of eight machines in consecutive memory locations. The supervisor wants to identify the machine with the highest production and determine whether the total production exceeds the daily target stored in another memory location, Design and implement an 8085 ALP to:
   - Calculate the total production.
   - Find the highest production value.
   - Compare the total with the target value.
   - Store a status flag indicating whether the target has been achieved.

2. An automated chemical plant records ten temperature readings. A reading is considered abnormal if it is greater than the predefined threshold stored in memory. The controller must count the number of abnormal readings and identify the highest recorded temperature. Write an 8085 ALP that
   - Scans all sensor readings.
   - Counts the number of abnormal readings.
   - Finds the maximum temperature.
   - Stores both results in memory.
3. An educational institution stores marks of ten students in memory. A student is considered successful if the mark is greater than or equal to the pass mark stored in memory. Develop an 8085 ALP to
   - Count the number of students who passed.
   - Find the highest mark.
   - Calculate the total marks obtained by all students.
   - Store all results in separate memory locations.

Code of Conduct

I Deepan Kumar P certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Deepan Kumar P

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly understands problem constraints, edge cases, and competitive requirements	Understands problem with minor gaps in constraints	Partial understanding; misses key constraints	Misinterprets problem or constraints
Algorithm	25%	Selects optimal algorithm with best time and space complexity for competitive constraints	Uses efficient algorithm with minor scope for improvement	Uses basic/inefficient approach	No clear algorithmic approach
Code Implementation	20%	Writes correct, efficient, and clean code that passes all constraints	Code works with minor errors or inefficiencies	Partially correct code; fails for some cases	Code does not execute or gives incorrect results
Competitive Performance	20%	Solves problem within time limits and handles large inputs effectively	Solves within acceptable time with minor delays	Struggles with time limits or larger inputs	Unable to solve within time constraints
Test Case Handling	15%	Handles all test cases including edge and hidden cases	Handles most test cases	Misses important edge cases	Fails on multiple test cases

Marks Evaluation:

Q. No	Problem Understanding (20)	Algorithm (25)	Code Implementation (20)	Competitive Performance (20)	Test case handling (15)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	

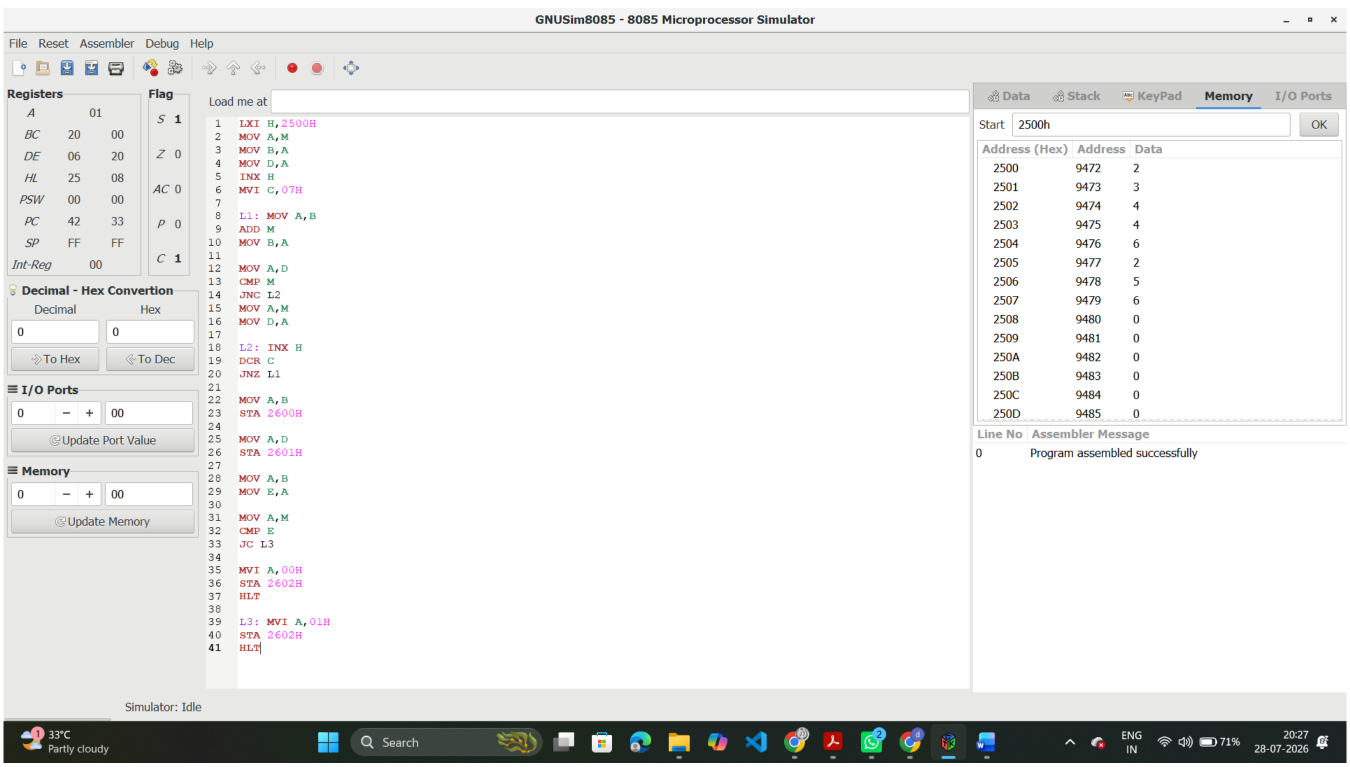
	/ 20	/ 25	/ 20	/ 20	/ 15	/ 100

Course Coordinator **Faculty Incharge**

1. A smart factory stores the hourly production of **8 machines** in consecutive memory locations.

Algorithm

- Load first production value.
- Initialize
    - Total = first value
    - Maximum = first value
- Read remaining 7 values.
- Add each value to Total.
- Compare each value with Maximum.
- Update Maximum if larger.
- Read Target.
- Compare Total with Target.
- If Total ≥ Target store 01.
- Else store 00.
- Store Total and Maximum.

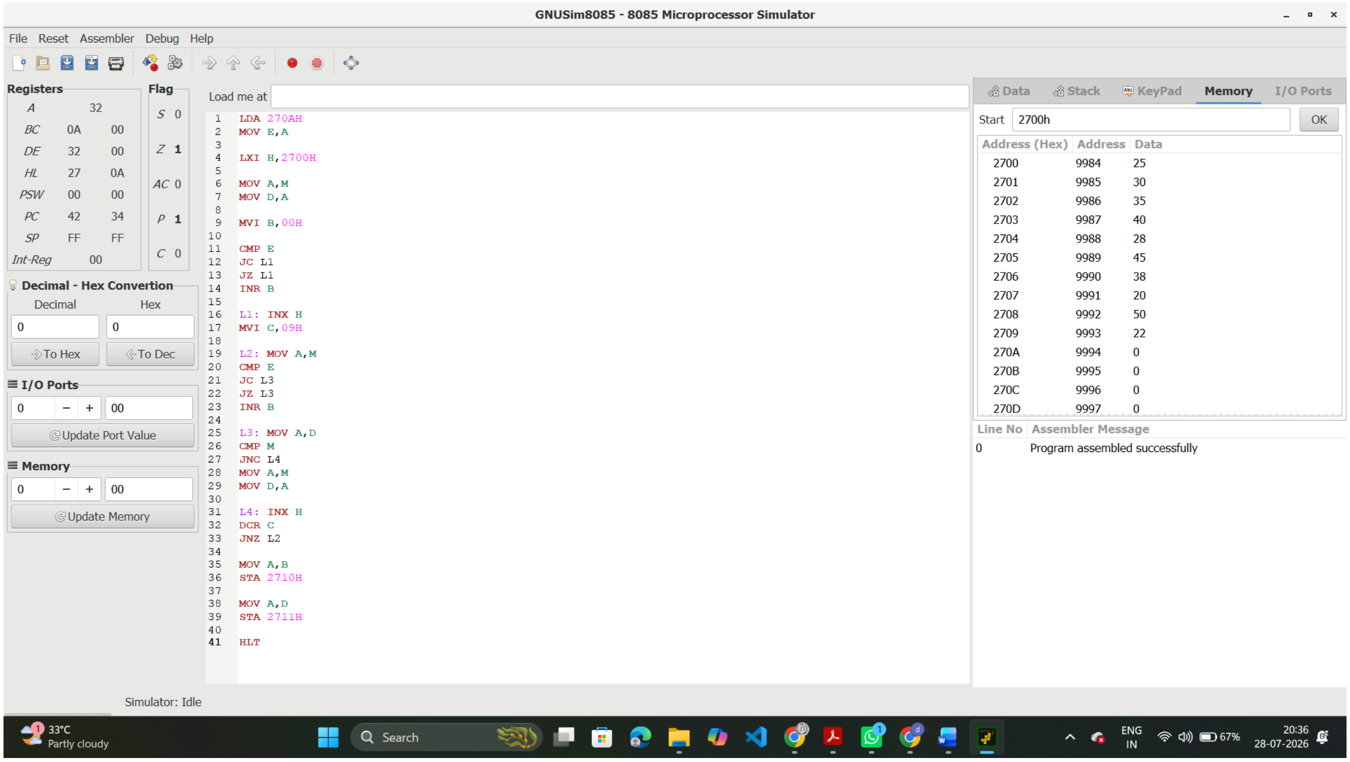


2. Temperature readings of **10 sensors** are stored.

    - **Algorithm**

- Read threshold.
- Read first temperature.
- Set Maximum.
- Count = 0.
- Compare each reading with threshold.
- If greater, increment count.
- Compare with Maximum.
- Update Maximum.

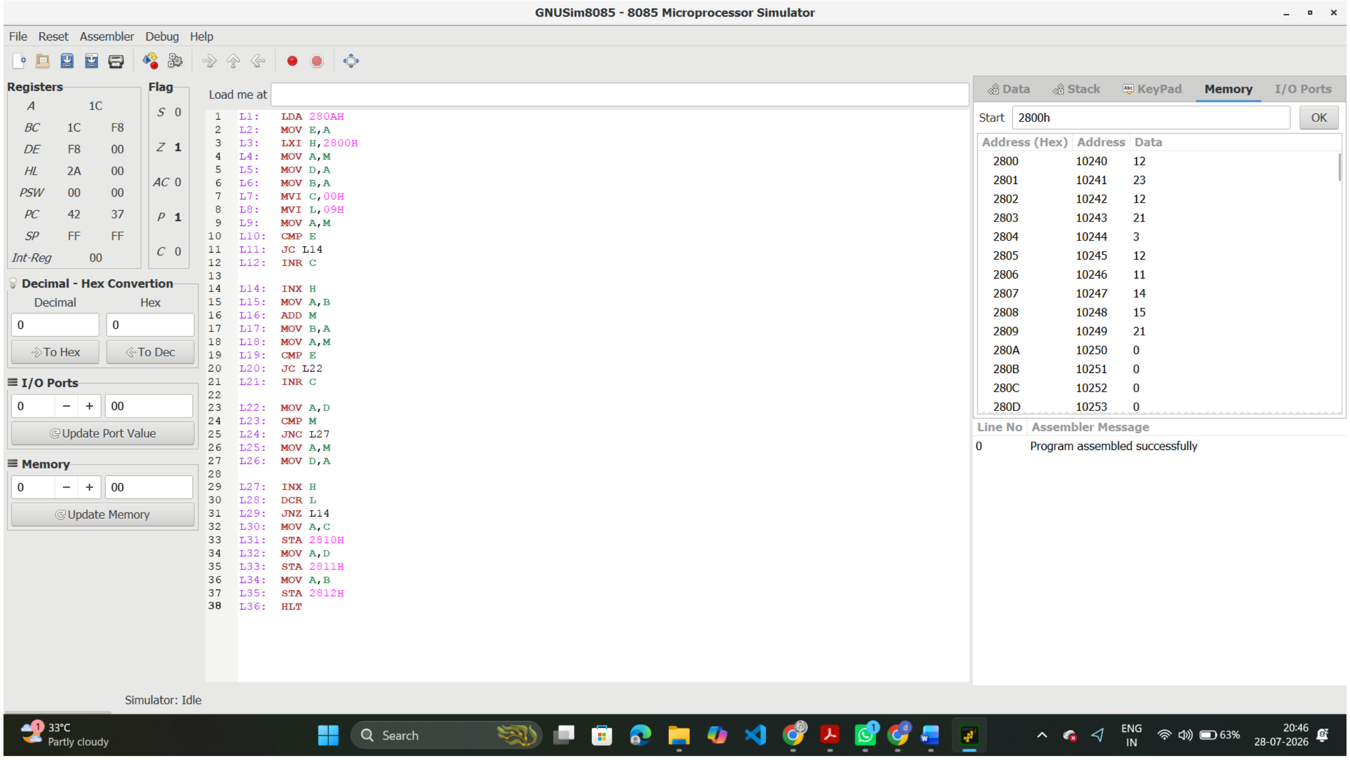
- Repeat for all 10 readings.
- Store Count and Maximum.



3. Marks of **10 students** are stored.

   <u>**Algorithm**</u>

- Read Pass Mark.
- Read first mark.
- Set
  - Highest = first mark
  - Total = first mark
- Count = 1 if first mark passes.
- Scan remaining marks.
- Add each mark to Total.
- Compare with Pass Mark.
- Increment pass count.
- Compare with Highest.
- Update Highest.
- Store Pass Count, Highest and Total.


GNUSim8085 - 8085 Microprocessor Simulator
File Reset Assembler Debug Help
Registers
A 1C
BC 1C F8
DE F8 00
HL 2A 00
PSW 00 00
PC 42 37
SP FF FF
Int-Reg 00
Flag
S 0
Z 1
AC 0
P 1
C 0
Decimal - Hex Convertion
Decimal Hex
To Hex To Dec
I/O Ports
Update Port Value
Memory
Update Memory
Load me at
L1: LDA 280AH
L2: MOV E,A
L3: LXI H,2800H
L4: MOV A,M
L5: MOV D,A
L6: MOV B,A
L7: MVI C,00H
L8: MVI L,09H
L9: MOV A,M
L10: CMP E
L11: JC L14
L12: INR C
L14: INX H
L15: MOV A,B
L16: ADD M
L17: MOV B,A
L18: MOV A,M
L19: CMP E
L20: JC L22
L21: INR C
L22: MOV A,D
L23: CMP M
L24: JNC L27
L25: MOV A,M
L26: MOV D,A
L27: INX H
L28: DCR L
L29: JNZ L14
L30: MOV A,C
L31: STA 2810H
L32: MOV A,D
L33: STA 2811H
L34: MOV A,B
L35: STA 2812H
L36: HLT
Data Stack KeyPad Memory I/O Ports
Start 2800h OK
Address (Hex) Address Data
2800 10240 12
2801 10241 23
2802 10242 12
2803 10243 21
2804 10244 3
2805 10245 12
2806 10246 11
2807 10247 14
2808 10248 15
2809 10249 21
280A 10250 0
280B 10251 0
280C 10252 0
280D 10253 0
Line No Assembler Message
0 Program assembled successfully
Simulator: Idle
33°C
Partly cloudy
Search
ENG
IN
63%
20:46
28-07-2026